Lab 5: Multiple Linear Regression

STAT 141, Week 5

Your name here

In this lab, you will

1. Review exploratory data analysis (EDA) techniques in R with multiple predictors
2. Assess linear regression models with multiple predictors
3. Interpret the estimated coefficients of multiple linear regression models
4. Compare and assess linear regression models

Add any collaborators here!

I collaborated with…

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

Run the chunk below to load the necessary packages.

```r
library(tidyverse)
library(moderndive)
```

Exercises

Exercise 1: Exploratory Data Analysis for Regression

We will study data on teacher salaries for the rest of this homework assignment. Our data is a sample of 100 post-secondary (i.e., college) teachers is collected from the 2010 American Community Survey Public Use Microdata Sample, as found in the `Lock5Data` R package. We are interested in exploring differences in salary based on gender (i.e., gender-based salary discrimination) among college teachers.

For this exercise, we will focus on the role of exploratory data analysis for linear regression models.

Load the dataset using the following code:

```
library(Lock5Data)
data("SalaryGender")
SalaryGender <- SalaryGender %>%
  select(Gender, Age, Salary) %>%
  mutate(Gender = ifelse(Gender == 0, "Female", "Male"))
```

The data includes three variables:

1. Gender: Coded as a binary "Female" or "Male". It is not clear whether this truly reflects the gender identity of participants, or something different like biological sex. (We might assume that no participants with other gender identities were included, which limits our ability to generalize to the population at large.)
2. Age: Age of participant, in years.
3. Salary: Annual salary in thousands of US dollars.

a. As stated above, we are interesting in exploring salary differences based on gender. It's important to start by thinking critically about the type of conclusions we can draw from this analysis, especially since we're analyzing data about a sensitive (and very important) topic. **What type of study is this data from, and what types of conclusions can be drawn as a result?** [Text only for this question]

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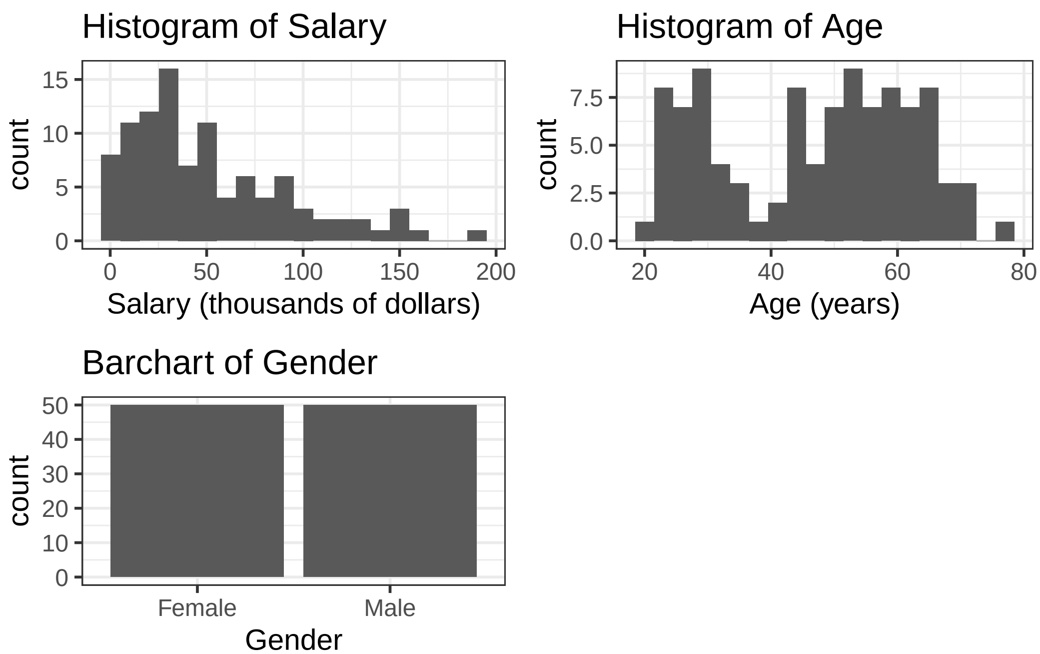
b. The chunk below creates exploratory visualizations of each variable: `Salary`, `Age`, and `Gender`. **Why is it important to perform this step before building a linear regression model? Based on the visualizations, give an insight about each variable.** [Text only for this question]

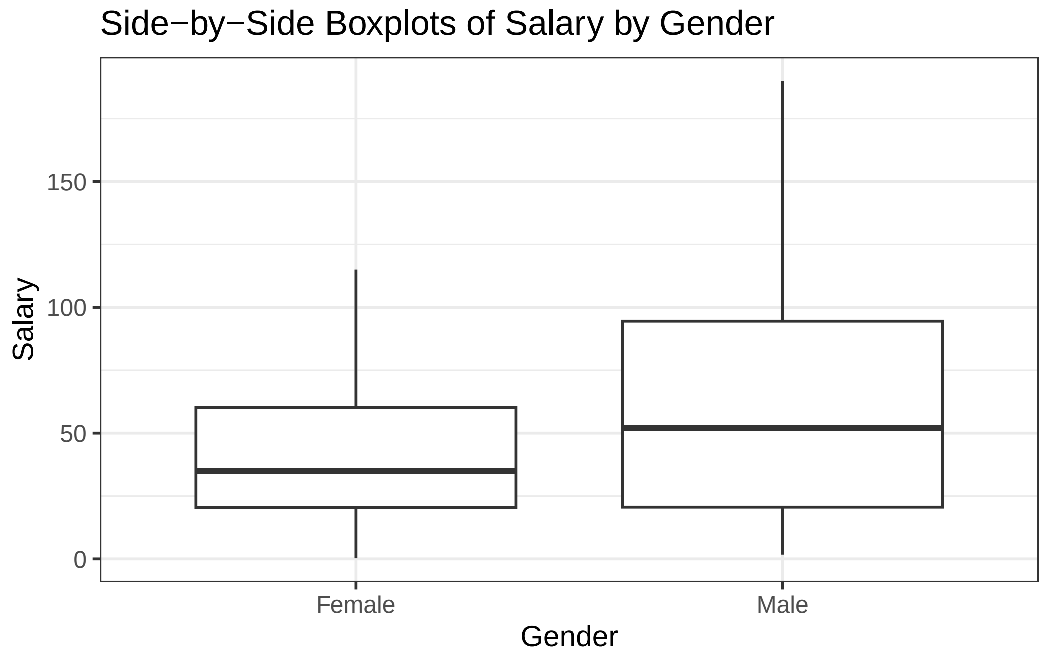
```
g1 <- ggplot(data = SalaryGender, aes(x = Salary)) + geom_histogram(bins = 20) +
  theme_bw() + labs(title = "Histogram of Salary", x = "Salary (thousands of dollars)")
g2 <- ggplot(data = SalaryGender, aes(x = Age)) + geom_histogram(bins = 20) +
  theme_bw() + labs(title = "Histogram of Age", x = "Age (years)")
g3 <- ggplot(data = SalaryGender, aes(x = Gender)) + geom_bar() +
```

```
  theme_bw() + labs(title = "Barchart of Gender")
gridExtra::grid.arrange(g1, g2, g3, nrow = 2)
```



c. The chunk below investigates the bivariate relationship between `Salary` and `Gender`. **What trends do you observe, and how are they relevant to building our linear model?** [Text only for this question]

```
ggplot(data = SalaryGender, aes(x = Gender, y = Salary)) + geom_boxplot() +
  theme_bw() + labs(title = "Side-by-Side Boxplots of Salary by Gender")
```



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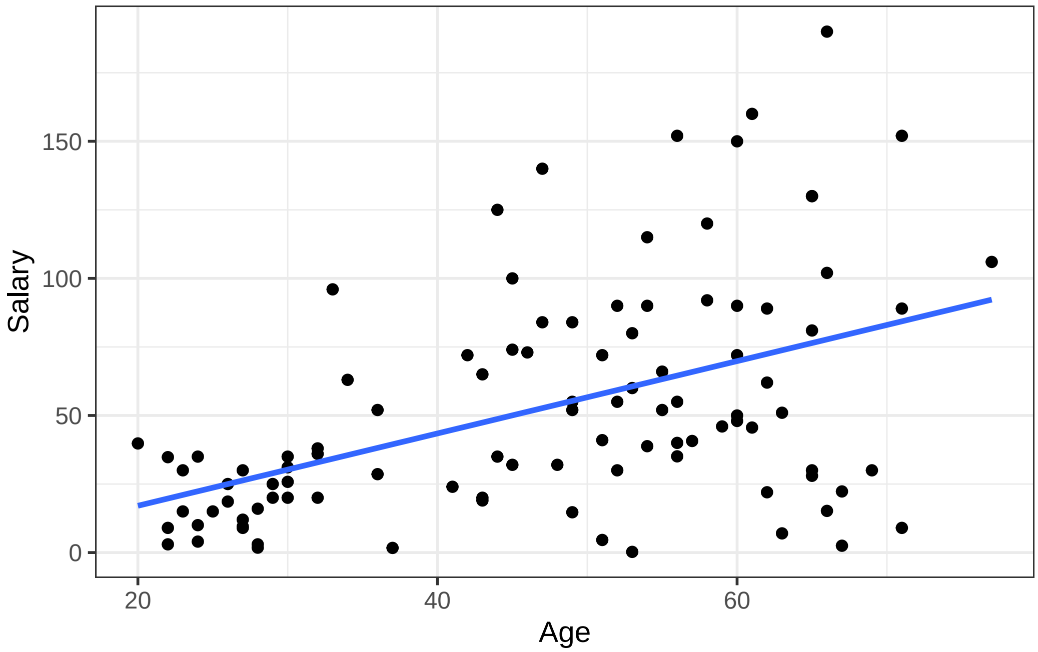
d. Assess the scatterplot produced by the chunk below. **Describe what the graph is showing.** [Text only for this question]

```
ggplot(SalaryGender, aes(x = Age, y = Salary)) + geom_point() +
  geom_smooth(method = "lm", se = F) + theme_bw()
```



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e. Calculate the correlation coefficient between `Salary` and `Age`, and **comment on the strength of the relationship**. [Code and text for this question]

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f. Now create a plot to investigate the bivariate relationship between `Age` and `Gender`. **Describe what your graph shows. Explain how insights from this graph and the graph from part (d) are relevant to building our linear model.** Keep in mind that our overall goal is to understand the relationship between salary and gender. [Code and text for this question]

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g. In a new chunk below, create a scatterplot of salary vs age, colored by gender. Add gender-specific trend lines by adding `+ geom_parallel_slopes(se = FALSE)`. **Comment on any trends observed.** [Code and text for this question]

Exercise 2: Salary Regression Model

In this exercise, you will use a regression model to investigate whether age explains the gap in salary between males and females in post-secondary education. (Note: Even *if* age explained the gap, **there is still a gap!** This would be an indication of structural differences for men and women in the US.)

a. The chunk below creates a linear model using the `SalaryGender` data frame. **Using the summary table, write the equation for the fitted model. Interpret each coefficient, and explain how their values validate your exploratory findings from Exercise 1.** [Only text needed for this question]

```
lm_salary_gender <- lm(Salary ~ Age + Gender, data = SalaryGender)
get_regression_table(lm_salary_gender)
```

```
# A tibble: 3 x 7
  term         estimate std_error statistic p_value lower_ci upper_ci
  <chr>           <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 intercept      -13.3     12.0       -1.10   0.272  -37.2      10.6
2 Age              1.24     0.244      5.06   0        0.751     1.72
3 Gender: Male    15.8      7.43       2.12   0.036    1.02     30.5
```

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b. Using your regression equation from part (a) of this exercise, calculate the model's prediction for the salary of a female teacher who is 30 years old. **Based on the exploratory visualizations in Exercise 1, is this an example of extrapolation?** [Code and text for this question]

Exercise 3: Assess Salary Regression Models

a. The model created for the previous exercise is just one example of a model that we could fit using the data set. In the chunk below, add a model named `m1` that predicts `Salary` using only `Gender`, and a model named `m2` that predicts `Salary` using only `Age`. [Only code for this question]

```r
# Create m1 here


# Create m2 here


# Recreate our previous model as m3 here
m3 <- lm(Salary ~ Age + Gender, data = SalaryGender)
```

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b. Which of the three models helps us best answer our original research question? **Consider both R^2 and the types of relationships each model captures in your argument.** Recall that you can find the R^2 of a model by using `get_regression_summaries()`. [Code and text for this question]

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c. To further investigate these models, let's work more closely with their predictions (aka fitted values) and residuals. Use `get_regression_points()` three separate times to grab a data frame with these sets of values from each of the models `m1`, `m2`, and `m3`. Assign these data frames (the output of `get_regression_points()` for each model) the names `m1_pts`, `m2_pts`, and `m3_pts`. [Code only for this question]

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d. Investigating when linear regression model assumptions hold is important to consider when selecting the best model(s) from a collection of possible models. **Remove eval=FALSE** at the top of the chunk below once you've finished 3(a), and run the chunk to assess the set of graphs shown. **Which LINE assumption(s) do these graphs relate to, and which model(s) satisfy it/them?** [Text only for this question]

```
# These next two commands are not usually needed, but here they help us compare
# the three different models directly within the a shard data frame (and
# then the same ggplot).
res <- bind_rows(m1_pts %>% mutate(model = "Salary ~ Gender") %>%
                   select(Salary, Salary_hat, residual, model),
                 m2_pts %>% mutate(model = "Salary ~ Age") %>%
                   select(Salary, Salary_hat, residual, model),
                 m3_pts %>% mutate(model = "Salary ~ Age + Gender") %>%
                   select(Salary, Salary_hat, residual, model))
res$model <- factor(res$model, levels = c("Salary ~ Gender",
                                          "Salary ~ Age",
                                          "Salary ~ Age + Gender"))

ggplot(res, aes(x = Salary_hat, y = residual)) +
  geom_hline(aes(yintercept = 0), lty = 2, color = "gray") +
  geom_point() +
  facet_wrap(~model) +  # This gets us one plot per model - usually we don't include it
  labs(title = "Residuals vs Fitted Values") +
  theme_classic()
```

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e. When you've finished 3(c), **remove eval=FALSE** at the top of the chunk below and assess the next set of diagnostic graphs shown. **Which LINE assumption(s) do these graphs relate to, and which model(s) satisfy it/them?** [Text only for this question]

```
ggplot(res, aes(x = residual)) +
  geom_histogram(bins = 20, color = "white") +
  facet_wrap(~model) +  # This gets us one plot per model - usually we don't include it
  labs(title = "Residual Histograms") +
  theme_classic()
```
